1. Assume $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{p-1} x_{ip-1} + \varepsilon_i,\ \varepsilon_i \sim i.i.d. N(0, \sigma^2)$. Let $y = (y_1, \ldots, y_n)^T, \varepsilon = (\varepsilon_1, \ldots, \varepsilon_n)^T, \beta = (\beta_0, \ldots, \beta_{p-1})^T, X = (1, x_1, \ldots, x_{p-1})$, where $x_j = (x_{1j}, \ldots, x_{nj}), j = 1, \ldots, p-1$.

   (a) Let the best estimate of β as the one which minimizes

$$\sum_{i=1}^{n} \varepsilon_i^2.$$

   Derive the least square estimate (OLS) $\hat{\beta}$.

   (b) Find the distribution of $\hat{\beta}$.

   (c) Write down the equation of R^2 and the estimate of σ^2.

2. The linear regression model for the savings data:

   Dataset background:

   Description: The savings data frame has 50 rows and 5 columns. The data is averaged over the period 1960-1970.

   The description of the variables:

name	Description
sr	savings rate - personal saving divided by disposable income
Pop15	percent population under age of 15
Pop75	percent population over age of 75
Dpi	per-capita disposable income in dollars
ddpi	percent growth rate of dpi

   Output from a multiple linear regression analysis of the model:

```
Call:
lm(formula = sr ~ pop15 + pop75 + dpi + ddpi, data = savings)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.2422 -2.6857 -0.2488  2.4280  9.7509 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) 28.5660865  7.3545161   3.884 0.000334 ***
pop15       -0.4611931  0.1446422     (1) 0.002603 ** 
pop75       -1.6914977  1.0835989  -1.561 0.125530    
dpi         -0.0003369  0.0009311  -0.362 0.719173    
ddpi         0.4096949  0.1961971   2.088 0.042471 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.803 on 45 degrees of freedom
Multiple R-squared:  0.3385,	Adjusted R-squared:  0.2797 
F-statistic: 5.756 on 4 and 45 DF,  p-value: 0.0007904
```

   (a) Find out the t-value (1). What is the distribution of (1)?

   (b) Find out $\hat{\sigma}^2$.

(c) Please explain the estimate of pop15.

(d) Write down the estimated linear regression model.

(e) Suppose we have a new data whose pop15 = 30, pop75=2, dpi=2000, and ddpi = 3. Use the model to predict the savings rate. Write down the predicted value.

(f) Are any of the predictors helpful in predicting the response? Please explain. (Hint: $H_0: \beta_{pop15} = \beta_{pop75} = \beta_{dpi} = \beta_{ddpi} = 0$)

(g) If we want to do the backward variable selection, which variable should we remove?

3. Assume $y_i \sim Bernoulli(p_i)$ for $\mathrm{i} = 1, \ldots, \mathrm{n}$, where $p_i = \Pr(y_i = 1|x_i)$.

The logistic regression model is defined as

$$\ln\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_i$$

(a) Show that the probability p_i can be expressed as the logistic function:

$$p_i = \frac{1}{1+e^{-(\beta_0+\beta_1 x_i)}}$$

(b) Instead of minimizing the sum of squared errors, logistic regression uses Maximum Likelihood Estimation (MLE). Write down the Log-likelihood function $l(\beta_0, \beta_1)$ for this model.

4. The wbca data frame contains 681 cases of potentially cancerous tumors. The objective is to predict whether a tumor is malignant based on various cell characteristics.

Variable descriptions:

name	Description
Class (Response)	1 if malignant, 0 if benign.
Adhes	Marginal Adhesion.
BNucl	Bare Nuclei.
Chrom	Bland Chromatin
Thick	Clump Thickness
USize	Uniformity of Cell Size

Output from the logistic regression model:

```
Call:
glm(formula = Class ~ ., family = "binomial", data = wbca)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 11.16678    1.41491   7.892 2.97e-15 ***
Adhes       -0.39681    0.13384  -2.965  0.00303 **
BNucl       -0.41478    0.10230  -4.055 5.02e-05 ***
Chrom       -0.56456    0.18728  -3.014  0.00257 **
Epith       -0.06440    0.16595  -0.388  0.69795
Mitos       -0.65713    0.36764  -1.787  0.07387 .
NNucl       -0.28659    0.12620  -2.271  0.02315 *
Thick       -0.62675    0.15890  -3.944 8.01e-05 ***
UShap       -0.28011    0.25235  -1.110  0.26699
USize        0.05718    0.23271   0.246  0.80589
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 881.388  on 680  degrees of freedom
Residual deviance:  89.464  on 671  degrees of freedom
AIC: 109.46

Number of Fisher Scoring iterations: 8
```

(a) Under the null hypothesis $H_0: \beta_{Adhes} = 0$. Write down the test statistic and its distribution.

(b) Calculate the Odds Ratio (OR) for BNucl. If the BNucl score increases by 1 unit, by what percentage does the odds of the tumor being malignant increase?

(c) Explain the estimate of USize. Is this predictor statistically significant at the $\alpha = 0.05$ level?

(d) Write down the estimated linear predictor equation for the log-odds: $\hat{\eta} = \log\left(\frac{\hat{p}}{1-\hat{p}}\right)$.

(e) Calculate the Likelihood Ratio Test (LRT) statistic to test the global null hypothesis ($H_0 = \beta_{Adhes} = \beta_{BNucl} = \cdots = \beta_{USize} = 0$)

(f) Suppose we perform backward variable selection using the R command drop1(). Based on the output, which variable is the best candidate for removal in the first step? Justify your choice?

```
> drop1(r)
Single term deletions

Model:
Class ~ Adhes + BNucl + Chrom + Epith + Mitos + NNucl + Thick +
    UShap + USize
       Df Deviance    AIC
<none>      89.464 109.46
Adhes   1   98.844 116.84
BNucl   1  109.000 127.00
Chrom   1   99.841 117.84
Epith   1   89.613 107.61
Mitos   1   93.551 111.55
NNucl   1   95.204 113.20
Thick   1  110.239 128.24
UShap   1   90.627 108.63
USize   1   89.523 107.52
```

(g) Suppose we evaluate the current model (prior to variable selection) using an independent dataset. The observed outcomes (y) and their corresponding predicted probabilities ($\hat{p}$) are shown in the table below:

Class (y)	1	0	0	0	1	0	1
$\hat{p}$	0.943	0.876	0.269	0.003	0.937	0.827	0.301

Using a classification cutoff of $c = 0.5$ (where $\hat{y} = 1$ if $\hat{p} \geq 0.5$. $p = 0.5$, calculate the sensitivity and specificity of the model.

(h) Based on the data provided in part (g), plot the ROC curve, and calculate the Area Under the Curve (AUC).

5. A case-control study was conducted to investigate the association between asbestos exposure (石綿暴露) and lung cancer (肺癌).

Researchers identified 200 lung cancer patients (Cases).
Researchers identified 400 healthy individuals (Controls).
Among the Cases, 60 had a history of asbestos exposure.
Among the Controls, 40 had a history of asbestos exposure.

(a) Construct the 2x2 Table based on the description above, complete the following contingency table:

	Lung cancer (case)	Healthy (control)	Total
Exposed			
Non-exposed			
Total	200	400	600

(b) Our goal is to estimate the odds ratio (OR) to understand how asbestos exposure affects the incidence of lung cancer.

$$\mathrm{OR} = \frac{\dfrac{\Pr(lung\ cancer|Exposed)}{1-\Pr(lung\ cancer|Exposed)}}{\dfrac{\Pr(lung\ cancer|Non-Exposed)}{1-\Pr(lung\ cancer|Non-Exposed)}}$$

To evaluate the OR in the above formula, do we need to estimate $\Pr(lung\ cancer|Exposed)$ and $\Pr(lung\ cancer|Non-Exposed)$ first? Is it possible to obtain these estimates from this specific dataset? Can we do that by this dataset? If not, explain how we can still determine the Odds Ratio (OR) and calculate $\widehat{\mathrm{OR}}$.